

Fock-selected topological dark matter and metastable dark energy

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ARTICLE INFO

Editor: Yasaman Farzan

Keywords:

Fock symmetry
Topological dark matter
Hopf soliton
Metastable dark energy
Matter overlap
Effective field theory

ABSTRACT

We construct and test a candidate-level covariant effective field theory in which the defining algebraic condition motivated by the hidden Fock symmetry of the bound Coulomb problem selects a topological dark-matter sector, while a resolved matter-overlap mediator activates a positive metastable dark-energy branch. The order parameter is a real antisymmetric internal $SO(4)$ tensor M_{ab} . Its Pfaffian condition $C_2 = \text{Pf } M = 0$ selects the decomposable fixed-radius locus, changing the vacuum manifold from S^5 to $S^2 \times S^2$ and thereby producing $\pi_3 = \mathbb{Z} \oplus \mathbb{Z}$. The same transition is dynamically linked, through the mediator χ , to activation of the metastable scalar branch. We supplement this algebraic construction with a topology-preserving three-dimensional stationary calculation of the $(N_+, N_-) = (1, 0)$ Hopf sector, a tangent- and charge-projected fluctuation analysis with no resolved negative physical mode in the tested sector, and a homogeneous trigger evolution that generates the overlap dynamically and crosses the mediator and scalar thresholds while remaining inside the stated local principal-stability window. The positive scalar branch has a calculable barrier, curvature mass, and semiclassical decay scaling. An illustrative reduced Newtonian-gauge closure for a dilute Hopfion population further exhibits a stable weak-coupling regime with sub-percent linear-growth deformations relative to its uncoupled reference; this is a candidate-level consistency check, not an observational fit. These results support a topological dark-matter candidate and an overlap-mediated metastable dark-energy candidate within the stated EFT. Relic production and abundance, unrestricted finite-stiffness and scattering dynamics, production-derived fluid coefficients, radiative naturalness, Boltzmann-complete evolution, and likelihood-level cosmological viability remain open.

1. Introduction

A joint dark-sector construction must solve two qualitatively different problems. It must provide localized or particle-like excitations capable of a nonrelativistic, approximately pressureless limit, and it must provide a smooth long-lived branch with negative pressure. If both are to arise from one organized transition rather than from two unrelated fluids, the organizing structure should have a concrete dynamical consequence: it should distinguish two phases, support a localized sector in the activated phase, and admit a mechanism by which matter can drive the transition. We therefore seek an internal algebraic selection rule whose presence changes the topology of the vacuum manifold rather than merely relabelling its degrees of freedom. The hidden Fock symmetry of the bound Coulomb problem supplies a particularly economical template [1–3]. Angular momentum L and the rescaled Runge–Lenz vector A combine into a real antisymmetric $SO(4)$ tensor M_{ab} , while the hydrogenic realization satisfies

$$L \cdot A = 0. \quad (1)$$

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<https://doi.org/10.1016/j.nuclphysb.2026.117639>

Received 22 May 2026; Received in revised form 15 August 2026; Accepted 20 August 2026

Available online 26 August 2026

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In four internal Euclidean dimensions this is the Pfaffian condition $\text{Pf } M = 0$ and hence precisely the decomposability condition for a two-form. At fixed norm, imposing it changes the available configurations from S^5 to the oriented Grassmannian

$$\text{Gr}_2^+(\mathbb{R}^4) \simeq S_+^2 \times S_-^2, \quad (2)$$

whose third homotopy group is $\mathbb{Z} \oplus \mathbb{Z}$. Removing the Pfaffian selection restores S^5 , for which $\pi_3(S^5) = 0$. We import the Fock relation because it supplies a concrete algebraic selection rule whose presence or absence changes the topology of the vacuum and therefore the existence of Hopf sectors. The Fock condition is load-bearing in the present construction. The $SO(4)$ used here is a global internal organizing symmetry of the effective theory. It is not gauged, and we do not claim that the Coulomb $SO(4)$ itself is a microscopic cosmological symmetry or that its dark-sector realization has been derived from a UV completion. Rather, the Fock relation is adopted as the defining structural postulate of the EFT because its low-energy consequences are concrete and ablatable. Its microscopic origin and the size of possible explicit $SO(4)$ -breaking operators remain open. Deformations that preserve the homotopy type of the activated target may alter the soliton energetics without eliminating the Hopf classification; sufficiently strong breaking that changes or lifts that target can remove the $\mathbb{Z} \oplus \mathbb{Z}$ sectors. To make the selection dynamical, we replace the original direct squared-overlap operator by a canonical mediator χ . Matter overlap can destabilize the mediator origin; the mediator then destabilizes the pre-transition scalar branch. Once the scalar reaches its positive metastable branch, the coefficient multiplying $(\text{Pf } M)^2$ becomes positive and selects the decomposable vacuum. Thus the same overlap-mediated transition organizes both the appearance of the Hopf-supporting topology and activation of the positive metastable branch. The resolved two-stage construction also gives explicit trigger thresholds and an analytic ghost/gradient-safe window for the reduced order-parameter fluctuation system. The load-bearing chain is

$$\left| \begin{array}{l} M_{ab} \longrightarrow Q_{ab} \longrightarrow \text{overlap-mediated transition} \\ \longrightarrow S^5 \longrightarrow S^2 \times S^2 \longrightarrow \text{Hopfion DM} + \text{metastable DE candidates.} \end{array} \right| \quad (3)$$

The word *candidate* is essential. We use it only after establishing an explicit action, a nontrivial stationary unit-Hopf configuration, local projected stability, a dynamically generated trigger trajectory, and controlled pressureless and negative-pressure limits. We reserve *viable model* for completion of relic production, self-interaction, full cosmological perturbation, and observational likelihood analyses.

1.1. Relation to existing constructions

The present construction is distinct from massive-spin-two and modified-gravity dark-sector models. The fundamental variables M_{ab} are spacetime scalars carrying internal $SO(4)$ indices; neither $Q_{ab}[M]$ nor the trace-free Hilbert tensor $S_{\mu\nu}[M]$ is an independent propagating spacetime spin-two field or an additional metric fluctuation. The dark-matter carrier is therefore a topological soliton of an internal order parameter rather than a massive tensor quantum. Consequently, the candidate does not rely on the Fierz–Pauli or nonlinear massive-gravity constraint structure. The model is also more specific than a generic Faddeev–Skyrme dark sector [4–8]. The target space supporting the Hopf charges is not postulated directly: it is selected by the hydrogenic Fock condition $\text{Pf } M = L \cdot A = 0$. Removing the corresponding selection term enlarges the fixed-radius vacuum from $S^2 \times S^2$ to S^5 and eliminates the nontrivial third homotopy group. Moreover, the metastable negative-pressure branch is not identified with the soliton or topological energy. It is supplied by an explicit scalar potential, while the matter-overlap mediator correlates activation of that branch with the topology-selecting transition. The novelty lies in this combined chain of Fock selection, topology change, Hopf-sector formation, and overlap-mediated metastable activation, rather than in any one ingredient taken separately. The paper is organized as follows. Section 2 proves the algebraic Fock selection and the conditional uniqueness of the symmetric trace-free composite. Section 3 defines the mediator-resolved EFT. Sections 4 and 5 derive the trigger and topology change. Sections 6–8 present the stationary Hopf solution and projected Hessian. Section 9 gives the reduced homogeneous trigger benchmark. Section 10 states the dark-matter and dark-energy candidate limits, Section 11 derives the model-specific reduced perturbative cosmology, and Section 12 separates completed results from the remaining viability gates.

2. Fock order parameter and the unique STF channel

2.1. Antisymmetric generator and invariants

Let $M_{ab} = -M_{ba}$, with $a, b = 1, \dots, 4$, transform in $\Lambda^2 \mathbb{R}^4$. In the Coulomb realization,

$$M_{ij} = \epsilon_{ijk} L_k, \quad M_{i4} = A_i. \quad (4)$$

The two independent polynomial invariants are

$$C_1 = \frac{1}{2} M_{ab} M_{ab} = L^2 + A^2, \quad (5)$$

$$C_2 = \frac{1}{8} \epsilon^{abcd} M_{ab} M_{cd} = \text{Pf } M = L \cdot A. \quad (6)$$

For a real antisymmetric 4×4 matrix, $\det M = (\text{Pf } M)^2$. A two-form in four dimensions is decomposable if and only if its Pfaffian vanishes; consequently $C_2 = 0$ implies $M = u \wedge v$. Introduce the self-dual and anti-self-dual pieces,

$$M_{\pm} = \frac{1}{2} (M \pm \star M), \quad (7)$$

which transform as $(1, 0)$ and $(0, 1)$ under $SO(4) \simeq SU(2)_+ \times SU(2)_-$. With $\rho_\pm^2 = \frac{1}{2} M_{\pm ab} M_\pm^{ab}$,

$$\rho_+^2 + \rho_-^2 = C_1, \quad C_2 = \frac{1}{2}(\rho_+^2 - \rho_-^2). \quad (8)$$

The Fock condition is therefore the balance condition $\rho_+ = \rho_-$.

2.2. Canonical quadratic tensor

Define

$$K_{ab} = M_{ac} M_b^c, \quad (9)$$

$$Q_{ab} = K_{ab} - \frac{1}{4} \delta_{ab} K_c^c = M_{ac} M_b^c - \frac{1}{4} \delta_{ab} M_{cd} M_{cd}. \quad (10)$$

The tensor Q is real, symmetric, trace-free, local, algebraic, and $SO(4)$ -equivariant.

Theorem 1 (conditional uniqueness). *The space of local, zero-derivative, real quadratic $SO(4)$ -equivariant maps*

$$\text{Sym}^2(\Lambda^2 V) \longrightarrow \text{STF}_2(V), \quad V \simeq \mathbb{R}^4, \quad (11)$$

is one-dimensional. Therefore Eq. (10) is unique up to overall normalization within this class.

Proof. Under $SO(4) \simeq SU(2)_+ \times SU(2)_-$,

$$\Lambda^2 V \simeq (1, 0) \oplus (0, 1), \quad \text{STF}_2(V) \simeq (1, 1). \quad (12)$$

Moreover,

$$\text{Sym}^2(\Lambda^2 V) = [(2, 0) \oplus (0, 0)] \oplus [(0, 2) \oplus (0, 0)] \oplus (1, 1). \quad (13)$$

The representation $(1, 1)$ occurs exactly once, and Eq. (10) is a nonzero map into it. It therefore spans the relevant Hom space. $\square$

The qualification is important: derivatives, higher powers of M , parity-odd structures, or additional internal tensors generate other descendants. The theorem concerns the lowest-order local real quadratic channel.

2.3. Exact Casimir identities

Every real antisymmetric 4×4 matrix can be brought by an $SO(4)$ rotation to

$$M = a e_1 \wedge e_2 + b e_3 \wedge e_4. \quad (14)$$

Then $C_1 = a^2 + b^2$, $C_2 = ab$, and the four eigenvalues of Q are $\pm(a^2 - b^2)/2$, each twice. Hence

$$Q^2 = \frac{1}{4}(C_1^2 - 4C_2^2)\mathbf{1}, \quad (15)$$

$$Q : Q = C_1^2 - 4C_2^2. \quad (16)$$

Since $2|C_2| \leq C_1$,

$$0 \leq Q : Q \leq C_1^2. \quad (17)$$

At fixed C_1 , the Fock locus saturates the upper bound. Equivalently,

$$\mu C_2^2 = \frac{\mu}{4}(C_1^2 - Q : Q). \quad (18)$$

The Pfaffian penalty is therefore an exact norm-gap term for the unique quadratic STF channel. On the nonzero decomposable locus, $K = C_1 P_\Pi$ for the oriented two-plane Π , and

$$Q = C_1 \left(P_\Pi - \frac{1}{2} \mathbf{1} \right). \quad (19)$$

The positive eigenspace reconstructs the unoriented plane and the norm reconstructs C_1 . The sign $M \rightarrow -M$ is not retained. This internal tensor is an order parameter, not a propagating spacetime spin-two particle.

3. Mediator-resolved covariant candidate EFT

We promote the six independent components of $M_{ab(x)}$ to real spacetime scalars carrying internal $SO(4)$ indices. The covariant action is

Table 1

Field content and principal operators of the four-dimensional EFT. “Positive” refers to the displayed two-derivative kinetic sign. The tensors Q_{ab} and $S_{\mu\nu}$ are composites, not additional propagating fields.

Object	Spacetime/internal character	Mass dim.	Dynamics, interactions, and role
M_{ab}	Six real spacetime scalars in $\Lambda^2 \mathbb{R}^4$ of global internal $SO(4)$	1	Positive kinetic term Z_M ; radial and Pfaffian potentials; first-derivative quartic $\mathcal{L}_4$; source of the Fock-selected Hopf sector and of $S_{\mu\nu}$.
$Q_{ab}[M]$	Internal symmetric trace-free composite	2	No independent kinetic term; the unique quadratic channel in Theorem 1 and an algebraic diagnostic of the Fock locus.
$S_{\mu\nu}[M]$	Spacetime symmetric trace-free Hilbert-stress composite	4	No independent degree of freedom; enters the matter portal $g\chi^2 S:T_m/\Lambda^6$.
χ	Real spacetime scalar, internal singlet	1	Positive canonical kinetic term; $m_\chi^2 \chi^2/2 + \eta \chi^4/4$ and $-\kappa \Xi \lambda^2/2$; resolves the matter-overlap trigger.
λ	Real spacetime scalar, internal singlet	1	Positive canonical kinetic term; $U(\lambda) + \mu(\lambda)C_2^2 - \kappa \Xi \lambda^2/2$; carries the positive metastable branch and activates Pfaffian stiffness.
$g_{\mu\nu}$	Spacetime metric	0	Einstein–Hilbert kinetic term; couples covariantly to all sectors.
ψ_m	Minimally coupled matter fields	varies by model	Dynamics specified by $\mathcal{L}_m$; their pre-portal Hilbert stress $T_m^{\mu\nu}$ enters the overlap.

$$I = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R + \mathcal{L}_m(g, \psi) - \frac{Z_M}{4} \nabla_\mu M_{ab} \nabla^\mu M_{ab} - \frac{1}{2} \nabla_\mu \lambda \nabla^\mu \lambda - \frac{1}{2} \nabla_\mu \chi \nabla^\mu \chi - V(M, \lambda, \chi) + \mathcal{L}_4[M] + \frac{g\chi^2}{\Lambda^6} S_{\mu\nu}[M] T_m^{\mu\nu} \right]. \quad (20)$$

We use metric signature $(-, +, +, +)$ and $g > 0$. The potential is

$$V(M, \lambda, \chi) = \frac{\alpha}{4} (C_1 - \sigma_0^2)^2 + \mu(\lambda) C_2^2 + U(\lambda) + \frac{1}{2} m_\chi^2 \chi^2 + \frac{\eta}{4} \chi^4 - \frac{\kappa}{2} \Xi(\chi) \lambda^2, \quad (21)$$

$$\Xi(\chi) = \frac{\chi^2}{1 + \chi^2/\chi_s^2}, \quad \mu(\lambda) = \mu_v \left[1 - e^{-(\lambda/f)^4/\chi_\mu^4} \right]. \quad (22)$$

We require $Z_M, \alpha, \mu_v, m_\chi^2, \eta, \kappa > 0$. The saturating Ξ makes the negative mixed quadratic term bounded in the χ direction, while the positive quartic in U bounds the λ direction. [Table 1](#) makes the independent fields, composites, and mass dimensions explicit. Natural units with $\hbar = c = 1$ are used throughout.

The composite dimensions are

$$[C_1] = [C_2] = [Q_{ab}] = 2, \quad [S_{\mu\nu}] = [T_{m,\mu\nu}] = 4. \quad (23)$$

All displayed EFT coefficients $Z_M, \alpha, \mu_v, \eta, \kappa, \zeta, g, a, b, c, x_\mu$ are dimensionless. The scales $\sigma_0, f, \chi_s, \Lambda, \Lambda_U, m_\chi$ have mass dimension one; M_{Pl} is the reduced Planck mass. Thus $g\chi^2 S:T_m/\Lambda^6$ has operator dimension four as required. The scalar self-potential is

$$U(\lambda) = \Lambda_U^4 \left(\frac{a}{2} x^2 - \frac{b}{3} x^3 + \frac{c}{4} x^4 \right), \quad x = \lambda/f, \quad (24)$$

where $a, b, c > 0$. Besides $x = 0$, stationary points exist for $b^2 > 4ac$ at

$$x_\pm = \frac{b \pm \sqrt{b^2 - 4ac}}{2c}. \quad (25)$$

The interval

$$\frac{2b^2}{9} < ac < \frac{b^2}{4} \quad (26)$$

gives a barrier at x_- and a positive local minimum at $x_v = x_+$. The benchmark shape $a = 0.24, b = c = 1$ has $x_v = 0.6$ and $U(x_v)/\Lambda_U^4 = 0.0036$. [Fig. 1](#) shows the dimensionless functions.

3.1. Variational spacetime STF tensor

Let I_M denote the M -dependent part of the action before the matter-overlap coupling and define its Hilbert stress tensor by

$$\Theta_{\mu\nu}^{(M)} = -\frac{2}{\sqrt{-g}} \frac{\delta I_M}{\delta g^{\mu\nu}}. \quad (27)$$

The spacetime tensor entering the overlap is its trace-free part,

$$S_{\mu\nu}[M] = \Theta_{\mu\nu}^{(M)} - \frac{1}{4} g_{\mu\nu} \Theta^{(M)}. \quad (28)$$

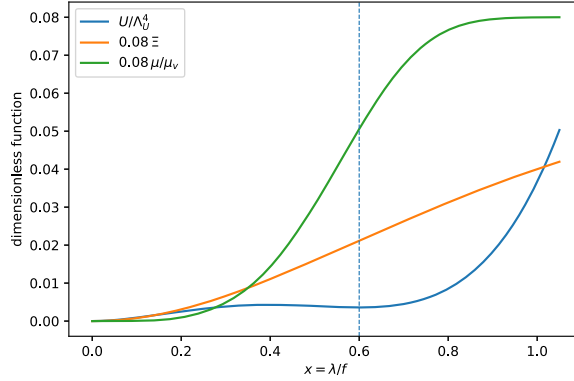


Fig. 1. Illustrative dimensionless scalar branch for $a = 0.24$, $b = c = 1$, together with $\Xi(\chi)$ at $\chi_s = f$ and the switching profile $\mu(\lambda)/\mu_v$ at the benchmark x_μ . The positive metastable minimum is at $\lambda/f = 0.6$; Ξ is rescaled for visibility.

This definition is covariant and fixed by variational differentiation; no soldering map between internal Euclidean and spacetime Lorentzian indices is introduced. At two-derivative order,

$$S_{\mu\nu}^{(2)} = \frac{Z_M}{2} \left(\nabla_\mu M_{ab} \nabla_\nu M_{ab} - \frac{1}{4} g_{\mu\nu} \nabla_\rho M_{ab} \nabla^\rho M_{ab} \right). \quad (29)$$

The potential contributes only a metric-proportional Hilbert stress and drops out of the trace-free part. The mediator in Eq. (20) replaces a direct high-dimension square of $S:T$. It keeps the fundamental Lagrangian second order in time derivatives and makes the background-dependent principal coefficients explicit. It does not, by itself, solve radiative naturalness or the full coupled matter-metric perturbation problem.

3.2. Quartic derivative stabilization

A polynomial first-derivative operator that reduces to a pair of Faddeev–Skyrme terms in the fixed-radius Fock vacuum is

$$\mathcal{L}_4 = \frac{\zeta}{16\sigma_0^4} \sum_{s=\pm} [(\nabla_\mu M_s \cdot \nabla_\nu M_s)(\nabla^\mu M_s \cdot \nabla^\nu M_s) - (\nabla_\mu M_s \cdot \nabla^\mu M_s)^2], \quad (30)$$

with $\zeta > 0$. On $M_s = (\sigma_0/\sqrt{2})\mathbf{n}_s$, $\mathbf{n}_s^2 = 1$, it is proportional to $-F_{s\mu\nu}F_s^{\mu\nu}$, where

$$F_{s\mu\nu} = \mathbf{n}_s \cdot (\nabla_\mu \mathbf{n}_s \times \nabla_\nu \mathbf{n}_s). \quad (31)$$

The Lagrangian contains first derivatives only. This avoids an Ostrogradsky degree of freedom, although hyperbolicity on arbitrary strong-gradient collision backgrounds remains a viability test.

3.3. Variational and conservation structure of the overlap

To avoid a self-referential stress tensor in the portal, define the matter tensor from the minimally coupled matter action before the portal is added,

$$T_{m,\mu\nu} \equiv -\frac{2}{\sqrt{-g}} \frac{\delta I_m}{\delta g^{\mu\nu}}, \quad I_m = \int d^4x \sqrt{-g} \mathcal{L}_m(g, \psi_m). \quad (32)$$

We assume that this matter EFT has no explicit curvature dependence. Write

$$F(\chi) \equiv \frac{g\chi^2}{\Lambda^6}, \quad X \equiv S_{\mu\nu}[M]T_m^{\mu\nu}, \quad I_p = \int d^4x \sqrt{-g} F X. \quad (33)$$

The portal contribution to the mediator equation and the full scalar equation are

$$\square\chi - m_\chi^2\chi - \eta\chi^3 + \frac{\kappa}{2}\Xi_{,\chi}\lambda^2 + \frac{2g\chi}{\Lambda^6}X = 0, \quad (34)$$

$$\square\lambda - U_{,\lambda} - \mu_{,\lambda}C_2^2 + \kappa\Xi(\chi)\lambda = 0. \quad (35)$$

Thus the portal has no direct λ variation, although the unrestricted theory retains the $\mu_{,\lambda}C_2^2$ coupling. The homogeneous system in Section 9 is a reduced overlap-generating proxy that does not evolve the Pfaffian degree of freedom; its scalar equation is correspondingly the $C_2 = 0$ reduction of Eq. (35), not the unrestricted six-component equation. For the two-derivative part of $S_{\mu\nu}$, define

$$\tilde{T}_m^{\mu\nu} \equiv T_m^{\mu\nu} - \frac{1}{4}T_m g^{\mu\nu}. \quad (36)$$

The portal contribution to the M_{ab} Euler–Lagrange equation is then

$$\mathcal{E}_{M_{ab}}^{p,(2)} = -Z_M \nabla_\nu \left[F(\chi) \tilde{T}_m^{\mu\nu} \nabla_\mu M_{ab} \right]. \quad (37)$$

The quartic Faddeev–Skyrme contribution can likewise be written in divergence form, as shown in [Appendix C](#). Because the displayed $\mathcal{L}_4$ depends only on first derivatives of the spacetime-scalar fields M_{ab} , the portal produces at most second derivatives of M within the stated EFT. Higher-derivative EFT descendants would require a separate principal-symbol analysis. For a matter field ψ_A , the species-independent form of the modified equation is

$$\mathcal{E}_A^{(m)} + F S_{\mu\nu} \frac{\delta T_m^{\mu\nu}}{\delta \psi_A} = 0, \quad (38)$$

where the functional derivative is converted to its Euler operator by integration by parts. No more explicit universal expression exists without specifying $\mathcal{L}_m$. Metric variation defines the portal stress

$$T_{\alpha\beta}^p = g_{\alpha\beta} F X - 2F \left[T_m^{\mu\nu} \frac{\delta S_{\mu\nu}}{\delta g^{\alpha\beta}} + S_{\mu\nu} \frac{\delta T_m^{\mu\nu}}{\delta g^{\alpha\beta}} \right], \quad (39)$$

with the two response kernels evaluated for the displayed M and chosen matter actions. Diffeomorphism invariance gives the on-shell identity

$$\nabla_\mu \left(T_m^{\mu\nu} + T_M^{\mu\nu} + T_\chi^{\mu\nu} + T_\lambda^{\mu\nu} + T_p^{\mu\nu} \right) = 0. \quad (40)$$

Equivalently, defining the species-dependent exchange current by

$$\nabla_\mu T_m^{\mu\nu} = J^\nu, \quad \nabla_\mu \left(T_M^{\mu\nu} + T_\chi^{\mu\nu} + T_\lambda^{\mu\nu} + T_p^{\mu\nu} \right) = -J^\nu, \quad (41)$$

makes explicit that the matter and dark subsystems need not be conserved separately. The exact J^ν is fixed only after the matter Lagrangian is chosen.

4. Two-stage overlap trigger and local stability window

4.1. Mediator and scalar thresholds

Write

$$X \equiv S_{\mu\nu} T_m^{\mu\nu}. \quad (42)$$

Near $\chi = \lambda = 0$, the effective mediator curvature is

$$m_{\chi,\text{eff}}^2 = m_\chi^2 - \frac{2gX}{\Lambda^6}. \quad (43)$$

Thus a positive overlap destabilizes the mediator origin when

$$X > X_{\chi,\text{crit}} = \frac{m_\chi^2 \Lambda^6}{2g}. \quad (44)$$

A fluctuation seed is required because the action has a $\chi \rightarrow -\chi$ symmetry and $\chi = 0$ is an exact classical trajectory. Once χ grows, the scalar curvature at the origin becomes

$$m_{\lambda,\text{eff}}^2 = U''(0) - \kappa \Xi(\chi) = \frac{a\Lambda_U^4}{f^2} - \kappa \Xi(\chi). \quad (45)$$

The scalar branch is tachyonic when

$$\Xi(\chi) > \frac{a\Lambda_U^4}{\kappa f^2}. \quad (46)$$

The scalar then rolls through the barrier into the positive local minimum. At tree level the quartic onset of $\mu(\lambda)$ leaves the leading curvature at the origin unchanged but makes $\mu(\lambda_v) > 0$ after activation. Its radiative status is discussed in [Section 10](#). The mechanism is hysteretic. When the matter overlap later decreases, the positive minimum of U remains separated by a barrier. For the benchmark $a = 0.24$, $b = c = 1$, the barrier and false minimum are at

$$x_{\text{top}} = 0.4, \quad x_v = 0.6, \quad (47)$$

with

$$U(x_{\text{top}}) = 4.2666667 \times 10^{-3} \Lambda_U^4, \quad U(x_v) = 3.6000000 \times 10^{-3} \Lambda_U^4. \quad (48)$$

The barrier above the false vacuum and the false-vacuum curvature mass are therefore

$$\Delta U_{\text{bar}} = 6.6666667 \times 10^{-4} \Lambda_U^4, \quad m_{\lambda,v}^2 \equiv U_{,\lambda\lambda}(\lambda_v) = 0.12 \frac{\Lambda_U^4}{f^2}. \quad (49)$$

This benchmark is not parametrically thin-wall. For the flat-space $O(4)$ bounce, the rescaling $\rho = \Lambda_U^2 r/f$ gives

$$B_4 = \left(\frac{f}{\Lambda_U} \right)^4 \hat{B}_4(a, b, c),$$

$$\hat{B}_4 = 2\pi^2 \int_0^\infty d\rho \rho^3 \left[\frac{1}{2} \left(\frac{dx}{d\rho} \right)^2 + u(x) - u(x_\nu) \right], \quad (50)$$

where $u = U/\Lambda_U^4$. Direct overshoot/undershoot shooting for the benchmark gives

$$x(0) = 0.00116696542914, \quad \hat{B}_4 = 1082.94756088. \quad (51)$$

The archived solution has a Euclidean virial relative defect 1.88×10^{-9} , and changing the integration truncation shifts $\hat{B}_4$ by 5.96×10^{-6} . This is a flat-space semiclassical diagnostic, not a gravitational Coleman–De Luccia lifetime; the prefactor and gravitational corrections must be included before quoting a lifetime in years.

4.2. Principal coefficients in a perfect-fluid background

The trigger must not rely on a background that already lies outside the derivative-stability domain. Consider one canonically normalized order-parameter fluctuation q in the rest frame of a perfect fluid,

$$T_m^{\mu\nu} = (\rho + p)u^\mu u^\nu + pg^{\mu\nu}. \quad (52)$$

For the scalar-like component of Eq. (29),

$$S_{\mu\nu} T_m^{\mu\nu} = (\rho + p) \left[\frac{3}{4} \dot{q}^2 + \frac{1}{4} (\nabla q)^2 \right] \quad (53)$$

up to the positive normalization absorbed into g . The quadratic principal Lagrangian is therefore

$$\mathcal{L}_{q,\text{prin}} = \frac{1}{2} K_M \dot{q}^2 - \frac{1}{2} G_M (\nabla q)^2, \quad (54)$$

with

$$K_M = 1 + \frac{3}{2}y, \quad G_M = 1 - \frac{1}{2}y, \quad c_M^2 = \frac{G_M}{K_M}, \quad y = \frac{g\chi^2(\rho + p)}{\Lambda^6}. \quad (55)$$

Consequently

$$\boxed{0 \leq y < 2} \quad (56)$$

is an analytic no-ghost, no-gradient-instability, and subluminal window for this order-parameter principal sector. For the actual reduced variables evolved in Section 9,

$$\mathbf{Y} = (q, \chi, \lambda)^T, \quad K_{\text{red}} = \text{diag}\left(1 + \frac{3}{2}y, 1, 1\right), \quad G_{\text{red}} = \text{diag}\left(1 - \frac{1}{2}y, 1, 1\right). \quad (57)$$

There are no mixed two-time-derivative or two-space-derivative entries in this reduced benchmark; the interactions enter its lower-derivative mass and source matrix. Hence Eq. (56) is the positivity condition for every eigenvalue of the complete kinetic and gradient matrices of the reduced q - χ - λ system actually evolved. It is not the principal matrix of the unrestricted matter–metric–six- M theory, whose species-dependent matter response and strong-gradient quartic sector remain separate viability calculations.

5. Topology change and trigger ablation

At $\lambda = 0$, $\mu = 0$. The radial potential fixes $C_1 = \sigma_0^2$ but leaves all six orientation components unrestricted,

$$\mathcal{V}_0 = \{M : C_1 = \sigma_0^2\} \simeq S^5, \quad \pi_3(S^5) = 0. \quad (58)$$

On the activated branch $\mu(\lambda_\nu) > 0$ imposes $C_2 = 0$, and hence

$$\rho_+^2 = \rho_-^2 = \frac{\sigma_0^2}{2}. \quad (59)$$

The vacuum is

$$\mathcal{V}_F \simeq S_+^2 \times S_-^2 \simeq \text{Gr}_2^+(\mathbb{R}^4), \quad \pi_3(\mathcal{V}_F) = \mathbb{Z} \oplus \mathbb{Z}. \quad (60)$$

Finite-energy maps are labelled by (N_+, N_-) . The transition is therefore a genuine change of the vacuum homotopy, not merely a reinterpretation of a pre-existing soliton sector. There is a direct ablation statement internal to the action and chosen basin of initial data. If $g = 0$, a matter overlap cannot lower $m_{\chi,\text{eff}}^2$; with initial data in the pre-Fock basin, χ and λ remain near zero, $\mu = 0$, and the vacuum remains S^5 . Alternatively, if $\kappa = 0$, the mediator may respond to overlap but cannot destabilize $\lambda = 0$, so the topology-selecting branch and positive metastable energy are not activated. The two couplings therefore carry distinct and indispensable

links in Eq. (3). Topology prevents continuous decay within $\mathcal{V}_F$ at finite energy. It does not forbid annihilation of opposite charges, radial-core excursions in the finite-stiffness theory, or gravitational collapse. Those processes require scattering and unrestricted-core simulations. At finite α and μ_v , exact homotopy protection on $\mathcal{V}_F$ becomes energetic persistence in the larger six-component field space. Along the direct vacuum-to-origin radial excursion, the local potential cost reaches

$$\Delta V_{\text{rad}}^{(0)} = \frac{\alpha \sigma_0^4}{4}. \quad (61)$$

At fixed radius, the exact inequality $2|C_2| \leq C_1 = \sigma_0^2$ gives the Pfaffian stiffness range

$$0 \leq \mu_v C_2^2 \leq \frac{\mu_v \sigma_0^4}{4}. \quad (62)$$

Using the soliton scales in Eq. (66), a core-size excursion has the characteristic estimates

$$\frac{E_{\text{rad}}^{\text{core}}}{M_H} \sim \frac{C_r}{4C_N} \frac{\alpha \zeta}{Z_M^2}, \quad \frac{E_{\text{Pf}}^{\text{core}}}{M_H} \sim \frac{C_p}{4C_N} \frac{\mu_v \zeta}{Z_M^2}, \quad (63)$$

where $C_r, C_p = \mathcal{O}(1)$ are path-dependent geometric factors. These are normal-stiffness and full-imbalance scales, not a calculation of the minimum-energy unwinding saddle. Determining that saddle and comparing it with collision and production energies requires an unrestricted finite-stiffness calculation.

6. Low-energy Hopf sector

On the final branch write

$$M_{\pm} = \frac{\sigma_0}{\sqrt{2}} \mathbf{n}_{\pm}, \quad \mathbf{n}_{\pm} \in S^2. \quad (64)$$

After a positive coordinate and energy rescaling, either decoupled sector has the dimensionless static functional

$$E[\mathbf{n}] = \int d^3x \left[\frac{1}{2} \partial_i \mathbf{n} \cdot \partial_i \mathbf{n} + \frac{1}{2} \sum_{i < j} \{ \mathbf{n} \cdot (\partial_i \mathbf{n} \times \partial_j \mathbf{n}) \}^2 \right]. \quad (65)$$

Under a spatial rescaling of size R , the two-derivative contribution scales as R and the quartic contribution as R^{-1} , producing a finite stationary scale [5,6,9]. In physical units,

$$R_H \sim \frac{\sqrt{\zeta}}{\sigma_0 \sqrt{Z_M}}, \quad M_H \sim \sigma_0 \sqrt{Z_M \zeta} C_N, \quad (66)$$

where C_N is a dimensionless charge-dependent number. The Vakulenko–Kapitanskii bound implies the characteristic scaling $E \geq c|N|^{3/4}$ [10]. For $\mathbf{n} \rightarrow \mathbf{n}_{\infty}$ at spatial infinity, compactified space is S^3 . Define

$$B_i = \frac{1}{2} \epsilon_{ijk} F_{jk}, \quad \nabla \times \mathbf{A} = \mathbf{B}, \quad \nabla \cdot \mathbf{A} = 0. \quad (67)$$

The Hopf invariant is [11,12]

$$Q_H = \frac{1}{16\pi^2} \int d^3x \mathbf{A} \cdot \mathbf{B}. \quad (68)$$

We take $\epsilon_{123} = +1$ and determine $\mathbf{A}$ by the Coulomb-gauge Fourier inversion in Eq. (68). With the orientation of the compactified initial map used below, the active + sector has $Q_H \simeq -1$; we define $N_+ \equiv -Q_H$ for this convention. Thus the computed unit solution is $(N_+, N_-) = (1, 0)$. The $(0, 1)$ sector follows from the $+$ $\leftrightarrow -$ symmetry, while joint, opposite-charge, and nonlinear charge-rearrangement sectors remain part of the scattering program.

6.1. Embedding in the full order-parameter theory

The numerical solve below is performed in the controlled fixed-radius, fixed-balance EFT obtained when the radial and Pfaffian-imbalance masses are heavier than the inverse soliton radius. The normal invariant curvatures are

$$\frac{\partial^2 V}{\partial C_1^2} = \frac{\alpha}{2} > 0, \quad \frac{\partial^2 V}{\partial C_2^2} = 2\mu(\lambda_v) > 0. \quad (69)$$

After fixing translations, global rotations, and Hopf charge, a non-degenerate reduced stationary point with a positive tangent gap persists for sufficiently large but finite positive normal stiffness by the implicit-function argument applied to the projected Euler–Lagrange map. The numerical Hessian in Section 8 supplies the required reduced tangent gap. This establishes a controlled finite-stiffness neighborhood, but it does not replace a future unrestricted six-component minimization at phenomenologically chosen finite α and μ .

Table 2

Topology-preserving stationary sequence. L , h , and E are dimensionless in the normalization of Eq. (65); no Fourier truncation beyond the N^3 grid is imposed. The virial column is $(E_2 - E_4)/E$.

N^3	L	h	E	Q_H	virial	constrained RMS
17^3	4	0.5000	281.3653	-0.999987	-8.20×10^{-2}	1.06×10^{-7}
21^3	4	0.4000	281.2711	-0.999727	-7.62×10^{-2}	9.41×10^{-6}
25^3	4	0.3333	281.2625	-0.999973	-7.58×10^{-2}	4.48×10^{-6}
33^3	8	0.5000	274.5448	-0.999931	-1.34×10^{-2}	5.08×10^{-6}

7. Three-dimensional stationary solution

7.1. Discretization and topology guard

We use a cubic domain $[-L, L]^3$ with N^3 sites and spacing $h = 2L/(N - 1)$. The boundary is fixed to $\mathbf{n}_\infty = (0, 0, -1)$, so the periodic extension used by the Fourier derivative is continuous. Spatial derivatives in both the energy and the Hopf functional are evaluated by the full-grid FFT convention with wave numbers $2\pi \text{fftfreq}(N, h)$; no additional Fourier-mode truncation or dealiasing is applied. At every optimizer step the interior field is projected pointwise to S^2 . A degree-one compactified Hopf map initializes the configuration. Because a finite lattice can change charge through unresolved configurations, we minimize the augmented functional

$$F = E + \ell(Q_H - Q_\star) + \frac{\beta}{2}(Q_H - Q_\star)^2, \quad Q_\star = -1, \quad (70)$$

with multiplier updates $\ell \rightarrow \ell + \beta(Q_H - Q_\star)$. The final stationary residual is measured after projecting the physical energy gradient onto the local S^2 tangents and subtracting its component normal to the constant- Q_H hypersurface. The archived augmented-Lagrangian runs use four outer updates with penalty $\beta = 2.0 \times 10^4$ and a strong-Wolfe L-BFGS inner solve with at most 120 iterations (180 function evaluations), gradient tolerance 10^{-10} , and change tolerance 10^{-14} .

7.2. Refinement and box-size behavior

Table 2 gives the fixed-box refinement sequence. The energy changes by 3.3×10^{-5} fractionally between the 21^3 and 25^3 runs, while $|Q_H|$ remains within 2.8×10^{-4} of unity. The finite $L = 4$ box compresses the configuration, as shown by the remaining virial mismatch. At fixed $h = 0.5$, enlarging to $L = 8$ yields a substantially improved virial balance.

For the $L = 8$ solution,

$$E_2 = 135.4296, \quad E_4 = 139.1152, \quad \frac{|E_2 - E_4|}{E} = 1.34 \times 10^{-2}. \quad (71)$$

The stationary charge is $Q_H = -0.999931$. Deterministic smooth tangent perturbations of the 25^3 , $L = 4$ checkpoint provide a local multistart check. The 1% and 2% RMS seeds return to configurations with $(E, Q_H) = (281.26228557, -0.999972096)$ and $(281.26225189, -0.999971797)$, constrained residuals 6.38×10^{-6} and 8.84×10^{-6} , and fractional energy shifts -6.51×10^{-7} and -7.71×10^{-7} , respectively. This supports robustness of the local unit-charge basin, not global uniqueness. Fig. 2 shows the central energy-density slice and the fixed-box refinement trend. These runs establish existence and local numerical stationarity in the unit-charge reduced sector; they are not yet a precision determination of C_N in Eq. (66).

8. Tangent- and charge-projected Hessian

A stationary configuration is not a dark-matter candidate if it possesses a negative physical fluctuation mode. At every interior site we construct an orthonormal tangent frame $(\mathbf{e}_1, \mathbf{e}_2)$ and parameterize unrestricted local tangent fluctuations by

$$\mathbf{n}(\mathbf{u}) = \frac{\mathbf{n}_0 + u_1 \mathbf{e}_1 + u_2 \mathbf{e}_2}{|\mathbf{n}_0 + u_1 \mathbf{e}_1 + u_2 \mathbf{e}_2|}. \quad (72)$$

For the 17^3 lattice this gives $2(15^3) = 6750$ tangent degrees of freedom. We differentiate the spectral energy by automatic differentiation, form Hessian-vector products for the Lagrange-multiplier functional

$$E - \alpha Q_H, \quad (73)$$

and project input and output vectors orthogonally to ∇Q_H . The lowest eigenpairs are computed iteratively. The archived 17^3 , $L = 4$ charge-projected raw spectrum is reproducible. Its first ten site-normalized eigenvalues are

$$\begin{aligned} &\{-4.4095 \times 10^{-9}, -8.2154 \times 10^{-15}, 7.4036 \times 10^{-5}, 7.4038 \times 10^{-5}, \\ &3.5950 \times 10^{-3}, 4.6971 \times 10^{-3}, 4.6971 \times 10^{-3}, \\ &8.4450 \times 10^{-2}, 8.6095 \times 10^{-2}, 8.6095 \times 10^{-2}\}. \end{aligned} \quad (74)$$

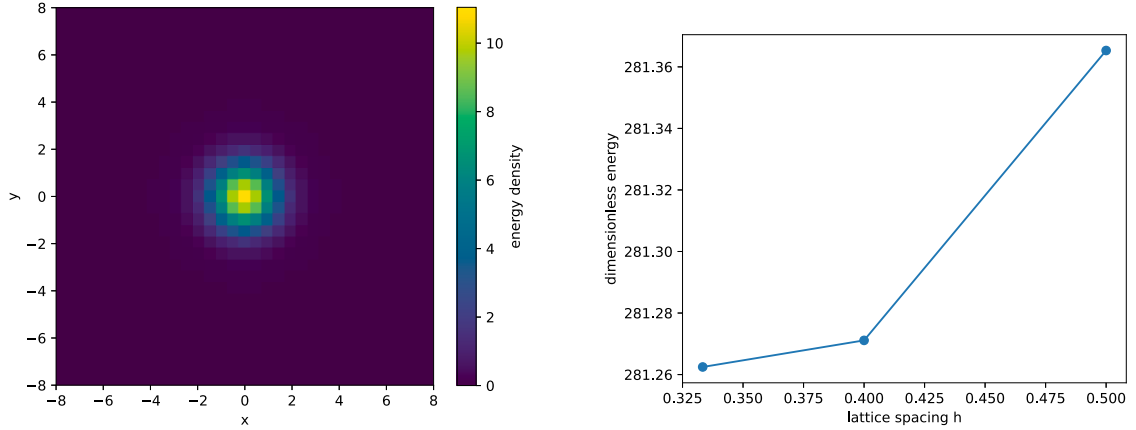


Fig. 2. Left: central dimensionless energy-density slice of the $N = 33$, $L = 8$, $h = 0.5$ stationary unit-Hopf configuration. Right: the $N = 17, 21, 25$ fixed- $L = 4$ refinement sequence in the normalization of Eq. (65). The energy approaches a stable lattice limit while the topology guard holds $Q_H \simeq -1$.

Table 3

Collective-deflated Hessian checks using the same constrained spectral operator. The first three rows form the fixed- $L = 4$ refinement sequence. The last two rows are resolved sine-band infrared checks on the enlarged box, not full-spectrum diagonalizations.

N^3	L	h	resolved subspace	$\lambda_{\min}^{\text{phys}}$
17^3	4	0.5000	full tangent, collective deflated	0.6750469263
21^3	4	0.4000	full tangent, collective deflated	0.6783182577
25^3	4	0.3333	full tangent, collective deflated	0.6759304079
33^3	8	0.5000	complete sine band $m_i \leq 4$ (128)	0.2815101819
33^3	8	0.5000	enlarged sine band $m_i \leq 5$ (250)	0.2632592918

Every eigenpair residual is $\mathcal{O}(10^{-14})$. These raw small values must not be identified with a non-collective shape gap. The projected operator contains a charge-projector null direction, an internal target rotation about the fixed vacuum axis, and finite-box translational and spatial-rotational pseudozero directions. In particular, the 7.40×10^{-5} pair belongs to this collective sector. We therefore construct the collective subspace from the charge gradient, one target rotation, three translations, and three spatial rotations, orthonormalize it in the unweighted lattice tangent inner product, and project it from the same constrained Hessian functional. Since the site inner product omits the volume element, the physical normalization is $\lambda_{\text{phys}} = \lambda_{\text{site}}/h^3$. Table 3 gives the first non-collective eigenvalue.

The three-point fixed-box fit linear in h^2 has intercept

$$\lambda_{\text{shape},1}^{\text{phys}}(h \rightarrow 0) = 0.6781839719 > 0, \quad (75)$$

with fit RMS residual 1.2541×10^{-3} . Only three resolutions enter, so we do not assign a formal continuum uncertainty. Neither enlarged-box band contains an eigenvalue below -10^{-10} after collective projection. They are resolved long-wavelength checks, not an unfiltered theorem for the full 59,582-dimensional 33^3 tangent space (see Fig. 3). This calculation is stronger than testing only invariant homogeneous curvatures: it permits all interior tangent deformations on the fixed- L lattices and resolves a separate large-box infrared subspace. Its limitations are finite volume and resolution, the fixed-radius reduction, and omission of coupled normal, gravitational, and mediator perturbations around a gravitating soliton.

9. Reduced homogeneous trigger benchmark

9.1. Reduced dynamical system

To test that the thresholds can be crossed by the model's own dynamics rather than by inserting a constant external X , consider a homogeneous order-parameter mode $q(t)$ in a pressureless matter bath. In dimensionless units set $f = \Lambda = 1$ and use

$$\rho(t) = \rho_0 e^{-\gamma t}, \quad X(t) = \frac{3}{4} \rho(t) \dot{q}^2. \quad (76)$$

The coupled equations are

$$K\ddot{q} + (3HK + \dot{K})\dot{q} + m_q^2 q = 0, \quad (77)$$

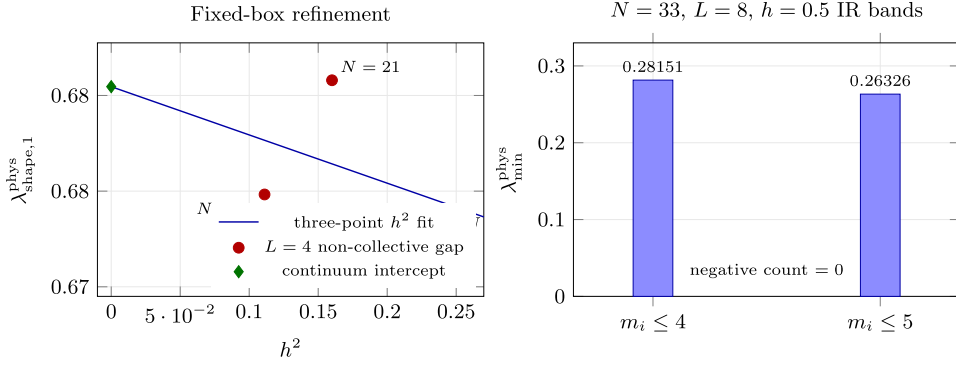


Fig. 3. Collective-deflated Hessian checks. Left: the first non-collective physically normalized eigenvalue for $N = 17, 21, 25$ at fixed $L = 4$, together with the three-point h^2 fit and its positive continuum intercept. Right: the $N = 33, L = 8, h = 0.5$ resolved long-wavelength sine-band minima for $m_i \leq 4$ and $m_i \leq 5$; these are band-restricted infrared checks rather than full-spectrum diagonalizations.

$$\ddot{\chi} + 3H\dot{\chi} + m_\chi^2\chi + \eta\chi^3 - \frac{\kappa}{2}\Xi_\chi\lambda^2 - \frac{3}{2}g\chi\rho\dot{q}^2 = 0, \quad (78)$$

$$\ddot{\lambda} + 3H\dot{\lambda} + U_{,\lambda} - \kappa\Xi(\chi)\lambda = 0, \quad (79)$$

where

$$K = 1 + \frac{3}{2}g\chi^2\rho. \quad (80)$$

The overlap in Eq. (76) is generated by $q(t)$ and feeds back into χ through Eq. (78). The matter bath and constant H are prescribed in this candidate benchmark; a closed FLRW fluid or lattice evolution is a later gate. We choose

$$\begin{aligned} m_q = 3, \quad m_\chi = 1, \quad \eta = 1, \quad \kappa = 0.55, \quad \Lambda_U^4 = 0.01, \quad \chi_s = 1, \\ a = 0.24, \quad b = c = 1, \quad g = 1, \quad \rho_0 = 0.28, \quad H = 0.035, \quad \gamma = 0.006, \end{aligned} \quad (81)$$

and initial data

$$q(0) = 1.3, \quad \dot{q}(0) = 0, \quad \chi(0) = 0.05, \quad \dot{\chi}(0) = 0, \quad \lambda(0) = 10^{-3}, \quad \dot{\lambda}(0) = 0. \quad (82)$$

The small nonzero mediator and scalar values represent generic fluctuations away from the exactly symmetric classical trajectory.

9.2. Transition and stability results

The dynamically generated overlap reaches

$$X_{\max} = 3.01077, \quad (83)$$

exceeding $X_{\chi,\text{crit}} = 0.5$ for the benchmark. The scalar curvature first becomes negative at

$$t_{\text{tr}} = 0.67398, \quad \min m_{\lambda,\text{eff}}^2 = -0.29388. \quad (84)$$

The fields reach

$$\chi_{\max} = 1.08060, \quad \lambda_{\max} = 0.69093, \quad \lambda(t = 300) = 0.60346. \quad (85)$$

The final value is close to the analytic metastable minimum $\lambda_v = 0.6$. By the end of the run, $X = 9.81 \times 10^{-15}$ and $\chi = -1.47 \times 10^{-7}$, while $m_{\lambda,\text{eff}}^2$ has returned positive. The scalar therefore remains trapped after the overlap and mediator have relaxed, explicitly realizing hysteresis in the benchmark. Throughout the run,

$$K_M \geq 1.00000, \quad G_M \geq 0.83987, \quad 0.56733 \leq c_M^2 \leq 1. \quad (86)$$

Thus the benchmark crosses the trigger while remaining inside the analytic principal-symbol window in Eq. (56). The baseline integration gives $\lambda(300) = 0.6034621611736$, $\min G_M = 0.8398719637$, and $\min c_M^2 = 0.5673338146$. Halving the maximum step or tightening or loosening the relative and absolute tolerances by a factor of ten leaves $\lambda(300)$ unchanged at approximately 10^{-10} absolute level and shifts the stability minima only at the few- 10^{-6} level. Fig. 4 shows the field evolution and stability coefficients. A local one-at-a-time sensitivity audit changes each of $g, \kappa, \rho_0, m_\chi$, and m_q independently by $\pm 5\%$. Every run crosses the scalar tachyonic condition, retains $G_M > 0$ and $c_M^2 > 0$, and reaches $\lambda_{\max} \geq 0.4$, the positive side of the barrier. The $m_q - 5\%$ run ends at $\lambda \simeq 0.4366$ rather than settling near 0.6 by $t = 300$, exposing a genuine capture boundary. These tests establish a local open neighborhood of the benchmark, not a global parameter scan.

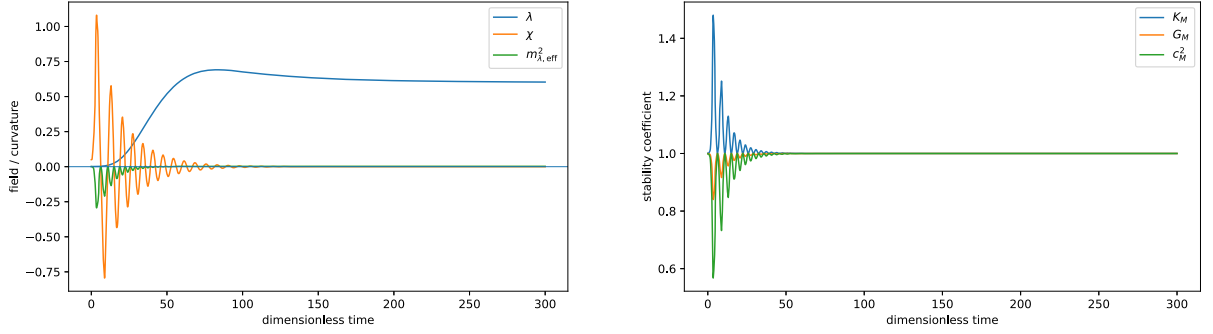


Fig. 4. Reduced homogeneous trigger benchmark with the dimensionless parameters and initial data of Eqs. (81) and (82). Left: mediator–scalar transition generated by the oscillating order-parameter proxy. Right: the eigenvalues K_M , G_M , and their ratio c_M^2 for the reduced principal matrices in Eq. (57).

For the reduced system, define

$$\mathcal{E} = \frac{1}{2}K\dot{q}^2 + \frac{1}{2}m_q^2q^2 + \frac{1}{2}\dot{\chi}^2 + \frac{1}{2}\dot{\lambda}^2 + V(\chi, \lambda). \quad (87)$$

The equations imply the work identity

$$\dot{\mathcal{E}} = -3H(K\dot{q}^2 + \dot{\chi}^2 + \dot{\lambda}^2) - \frac{3}{4}g\chi^2\dot{\rho}\dot{q}^2. \quad (88)$$

The numerical open-system work-balance residual of Eq. (88), normalized by the mean absolute right-hand side, is 2.07×10^{-3} . This is not a closed energy-conservation test: the last term is work performed by the explicitly time-dependent prescribed bath. In a closed cosmological model it is replaced by the matter/portal exchange in Eqs. (40)–(41). The present bath is used only to demonstrate a self-generated overlap trajectory and stable threshold crossing.

10. Dark-matter and dark-energy candidate limits

10.1. Hopfion population

Suppose the topology-changing transition produces a dilute population of stable or sufficiently long-lived Hopf solitons, and suppose their typical kinetic energy is small compared with M_H . If number-changing interactions are slow,

$$\dot{n}_H + 3Hn_H \simeq 0. \quad (89)$$

Then

$$\rho_H = M_H n_H + \mathcal{O}(n_H T_H), \quad p_H \simeq n_H T_H \ll \rho_H, \quad (90)$$

and hence

$$\rho_H \propto a^{-3}, \quad w_H \simeq 0. \quad (91)$$

The stationary solution and Hessian establish the localized carrier required for this limit. They do not determine the produced number density, velocity distribution, annihilation rate, or observed abundance. The four tangent Goldstone coordinates of $S^2 \times S^2$ can also carry relativistic radiation and mediate soliton interactions. A viable cosmology must show that residual waves, isocurvature, and long-range effects remain acceptable. These are not hidden by the topological argument.

10.2. Metastable scalar branch

For a scalar trapped near λ_v ,

$$\rho_\lambda = \frac{1}{2}\dot{\lambda}^2 + U_{\text{eff}}(\lambda), \quad p_\lambda = \frac{1}{2}\dot{\lambda}^2 - U_{\text{eff}}(\lambda). \quad (92)$$

When $\dot{\lambda}^2 \ll U_{\text{eff}}(\lambda_v)$,

$$w_\lambda \simeq -1. \quad (93)$$

The benchmark demonstrates capture near the positive metastable minimum. Matching its absolute value to

$$U_{\text{eff}}(\lambda_v) = 3M_{\text{pl}}^2 H_0^2 \Omega_{\text{DE},0} \quad (94)$$

is still a parameter condition. The tree-level mass $m_{\lambda,v}$ in Eq. (49) is the curvature of U at the false minimum; it is distinct from the effective final-branch mass $m_{\lambda,f}$ used in the reduced cosmology, which may include density and portal corrections.

Radiative status. The exact $\chi \rightarrow -\chi$ symmetry forbids odd- χ operators and protects the absence of a mediator tadpole; global internal $SO(4)$ restricts the M -sector counterterms to invariant combinations. The limit $g \rightarrow 0$ removes the matter-overlap portal. The limit $\kappa \rightarrow 0$ removes the direct χ - λ trigger, but it does not fully decouple λ from M because $\mu(\lambda)C_2^2$ remains. These structures do not protect the absolute dark-energy scale or the light scalar curvature. The cubic term in U precludes a $\lambda \rightarrow -\lambda$ symmetry, and the potential precludes a shift symmetry. Consequently the tree-level quartic onset $\mu(\lambda) \propto \lambda^4 + \dots$ is not enforced by the stated symmetries: lower-order $\lambda^n C_2^2$ counterterms are allowed. We therefore treat $\mu_{\text{eff}}(0) = 0$ and the switching hierarchy as renormalized EFT matching conditions, not symmetry-protected predictions. The evenness of $\Xi(\chi)$ is protected, but its exact saturating functional form is not. The absolute value $U(\lambda_v)$, the ultra-light scalar scale, and the matching condition require radiative tuning or an additional UV protection mechanism. This naturalness limitation is separate from the candidate-level result that the stated renormalized EFT possesses a dynamically accessible metastable positive branch.

10.3. Background accounting

A candidate late-time Friedmann system has the schematic form

$$3M_{\text{Pl}}^2 H^2 = \rho_r + \rho_b + \rho_H + \rho_\lambda + \rho_{\text{waves}} + \rho_{\text{portal}}, \quad (95)$$

$$-2M_{\text{Pl}}^2 \dot{H} = \frac{4}{3}\rho_r + \rho_b + \rho_H + \dot{\lambda}^2 + \rho_{\text{waves}} + p_{\text{waves}} + \rho_{\text{portal}} + p_{\text{portal}}. \quad (96)$$

Because Eq. (20) is diffeomorphism invariant, the sum of the matter, M , χ , λ , and interaction stresses is covariantly conserved. Energy transfer between sectors must therefore be followed explicitly in an expanding-lattice production calculation rather than assigned by hand.

11. Model-specific reduced perturbative cosmology

This section is a candidate-level internal-consistency test, not a Boltzmann prediction or an observational fit. The stationary and trigger analyses test local field-theory consistency, but a model-specific description is also needed for how the final dark-sector branch perturbs an FLRW spacetime. We therefore construct a reduced long-wavelength closure appropriate to the actual degrees of freedom of the present model. It is deliberately narrower than a Boltzmann-complete treatment: the Hopfion population is coarse-grained as a dilute fluid, the post-transition mediator is expanded around its symmetric origin, and residual Goldstone waves, formation shot noise, and primordial isocurvature are neglected. These assumptions define the domain in which the following equations are applicable.

11.1. Final-branch degrees of freedom

After the trigger has relaxed, take

$$\bar{\chi} = 0, \quad \bar{\lambda} \simeq \lambda_v, \quad m_{\chi,f}^2 \equiv m_\chi^2 - \kappa \lambda_v^2 > 0, \quad m_{\lambda,f}^2 \equiv U_{\text{eff},\lambda\lambda}(\lambda_v) > 0. \quad (97)$$

Because the action is even under $\chi \rightarrow -\chi$, the mediator has no linear mixing with matter, λ , or the Hopfion fluid at $\bar{\chi} = 0$ and vanishing background overlap. Its linear perturbation therefore obeys

$$\delta\chi'' + 2H\delta\chi' + (k^2 + a^2 m_{\chi,f}^2)\delta\chi = 0, \quad (98)$$

where primes denote conformal-time derivatives and $H = a'/a$. Eq. (98) is stable when Eq. (97) holds and may consistently be left unexcited in the reduced adiabatic sector. At finite radial and Pfaffian stiffness, the soliton mass can inherit a weak dependence on the branch scalar through $\mu(\lambda)$ and the normal-mode response. Define the finite-stiffness Wilson coefficient

$$\beta_H \equiv M_{\text{Pl}} \left. \frac{\partial \ln M_H(\lambda)}{\partial \lambda} \right|_{\lambda_v}. \quad (99)$$

In the strict fixed-radius Faddeev-Skyrme limit used for the stationary calculation, $\beta_H \rightarrow 0$ at leading order. A nonzero small value parameterizes the first model-specific correction that an unrestricted finite-stiffness soliton calculation must determine. Conservation of Hopfion number gives the reduced background equations

$$\bar{\rho}'_H + 3H\bar{\rho}_H = \frac{\beta_H}{M_{\text{Pl}}} \bar{\lambda}' \bar{\rho}_H, \quad (100)$$

$$\bar{\lambda}'' + 2H\bar{\lambda}' + a^2 U_{\text{eff},\lambda\lambda} = -a^2 \frac{\beta_H}{M_{\text{Pl}}} \bar{\rho}_H. \quad (101)$$

For $|\beta_H| \ll 1$ and $m_{\lambda,f} \gtrsim H$, the scalar follows the nearby density-dependent minimum and the background remains a small deformation of the candidate limit in Section 10.

11.2. Newtonian-gauge closure

Use conformal Newtonian gauge [13],

$$ds^2 = a^2 [-(1 + 2\Psi)d\tau^2 + (1 - 2\Phi)d\mathbf{x}^2]. \quad (102)$$

Let $\Delta_H \equiv \delta n_H / \bar{n}_H$ be the Hopfion number-density contrast and θ_H its velocity divergence. To linear order,

$$\delta_H \equiv \frac{\delta \rho_H}{\bar{\rho}_H} = \Delta_H + \frac{\beta_H}{M_{\text{Pl}}} \delta \lambda. \quad (103)$$

Number conservation and the scalar force generated by $M_H(\lambda)$ give the reduced fluid equations [14]

$$\Delta'_H = -\theta_H + 3\Phi', \quad (104)$$

$$\theta'_H + \left(H + \frac{\beta_H}{M_{\text{Pl}}} \bar{\lambda}' \right) \theta_H = k^2 \left(\Psi + \frac{\beta_H}{M_{\text{Pl}}} \delta \lambda + c_H^2 \Delta_H - \sigma_H \right). \quad (105)$$

The scalar perturbation satisfies

$$\delta \lambda'' + 2H \delta \lambda' + (k^2 + a^2 m_{\lambda,f}^2) \delta \lambda = -a^2 \frac{\beta_H \bar{\rho}_H}{M_{\text{Pl}}} \Delta_H + S_{\text{met}}, \quad (106)$$

where S_{met} contains the standard terms proportional to $\bar{\lambda}' \Phi'$, $U_{\text{eff},\lambda} \Psi$, and the small density-dependent displacement of the scalar minimum. These terms vanish or are Hubble-suppressed on the trapped slow-roll branch used below. Visible baryons remain minimally coupled,

$$\delta'_b = -\theta_b + 3\Phi', \quad \theta'_b + H\theta_b = k^2 \Psi. \quad (107)$$

The metric potentials are closed by the linear Einstein equations, including

$$k^2 \Phi + 3H(\Phi' + H\Psi) = -4\pi G a^2 \delta \rho_{\text{tot}}, \quad (108)$$

$$k^2(\Phi - \Psi) = 12\pi G a^2 (\bar{\rho}_H + \bar{p}_H) \sigma_H + \text{residual-wave stress}. \quad (109)$$

The long-wavelength properties of the soliton ensemble enter through a derivative expansion,

$$c_H^2(k, a) = c_v^2(a) + c_2 \left(\frac{k R_H}{a} \right)^2 + \mathcal{O}[(k R_H/a)^4], \quad (110)$$

$$\sigma_H(k, a) = c_\pi \left(\frac{k R_H}{a} \right)^2 \Delta_H + \mathcal{O}[(k R_H/a)^4]. \quad (111)$$

Here c_v^2 is the centre-of-mass velocity dispersion and c_2, c_π are finite-size response coefficients. Eqs. (110)–(111) are valid for a dilute population on scales $k R_H/a \ll 1$; collision simulations and the production-generated phase-space distribution are required to calculate their coefficients rather than treating them as reduced EFT data.

Status of the reduced coefficients. The three quantities entering the Hopfion closure have different microscopic status. The scalar charge β_H is a finite-stiffness Wilson coefficient. It vanishes at leading order in the strict fixed-radius/fixed-balance Faddeev–Skyrme limit, where M_H is independent of λ , and becomes nonzero only through the response of the normal soliton modes to $\mu(\lambda)$. The benchmark value $\beta_H = 0.02$ is therefore illustrative rather than predicted. The sound speed $c_H^2 = c_v^2 + c_2(k R_H/a)^2 + \dots$ is set by the centre-of-mass velocity dispersion and finite-core response. Thus $c_H^2 \ll 1$ is expected for a cold nonrelativistic population, but its numerical value must be obtained from production and kinetic evolution; $c_H^2 = 10^{-9}$ is an illustrative cold-population choice. Likewise, $\sigma_H = c_\pi(k R_H/a)^2 \Delta_H + \dots$ is derivative suppressed. We set $\sigma_H = 0$ only as the leading isotropic pressureless closure, not as a microscopic prediction. The benchmark therefore demonstrates the existence of a stable weak-coupling perturbative regime rather than predicting these closure coefficients.

11.3. Stability and quasi-static growth

The reduced scalar sector is free of a short-wavelength ghost or gradient instability when

$$\bar{\rho}_H + \bar{p}_H > 0, \quad 0 \leq c_H^2 \leq 1, \quad m_{\lambda,f}^2 > 0, \quad m_{\chi,f}^2 > 0. \quad (112)$$

The canonical $\delta \lambda$ and $\delta \chi$ kinetic terms are positive, while the Hopfion kinetic coefficient is its positive enthalpy. In the sub-horizon, slowly varying final branch, Eq. (106) gives

$$\delta \lambda \simeq -\frac{a^2 \beta_H \bar{\rho}_H}{M_{\text{Pl}}(k^2 + a^2 m_{\lambda,f}^2)} \Delta_H. \quad (113)$$

The scalar therefore changes the attractive Hopfion–Hopfion force by the non-negative fraction

$$\epsilon_H(k, a) \equiv \frac{G_{HH}}{G} - 1 = \frac{2\beta_H^2 k^2}{k^2 + a^2 m_{\lambda,f}^2}. \quad (114)$$

It does not directly change the baryon–baryon force. With $N = \ln a$, $\bar{\lambda}' \simeq 0$, $\sigma_H = 0$, and the quasi-static closure, the growth system becomes

$$\frac{d^2 D_H}{dN^2} + \left(2 + \frac{d \ln H}{dN} \right) \frac{d D_H}{dN} + \frac{c_H^2 k^2}{a^2 H^2} D_H = \frac{3}{2} [\Omega_b D_b + \Omega_H (1 + \epsilon_H) D_H], \quad (115)$$

Table 4

Illustrative reduced perturbation benchmark with $\beta_H = 0.02$, $m_{\lambda,f} = 5H_0$, $c_H^2 = 10^{-9}$, and $\sigma_H = 0$. The changes are relative to the uncoupled pressureless system on the same background and are not an observational fit.

$k/[h \text{ Mpc}^{-1}]$	$\epsilon_H(z=0)$	$\Delta D_H/D_H$	$\Delta D_b/D_b$
0.01	7.7835×10^{-4}	+0.2213%	+0.1548%
0.10	7.9978×10^{-4}	+0.2150%	+0.1531%
0.20	7.9994×10^{-4}	+0.1952%	+0.1481%

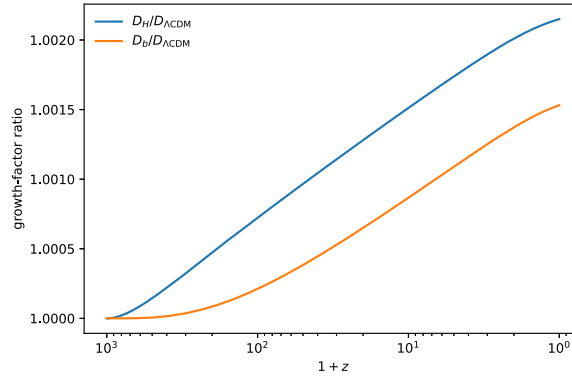


Fig. 5. Reduced growth-factor ratios at $k = 0.1 h \text{ Mpc}^{-1}$ for the illustrative closure choices in Eq. (118). The Hopfion and baryon sectors remain sub-percent deformations of the uncoupled reference history; this is a candidate-level consistency check, not an observational fit.

$$\frac{d^2 D_b}{dN^2} + \left(2 + \frac{d \ln H}{dN}\right) \frac{d D_b}{dN} = \frac{3}{2} (\Omega_b D_b + \Omega_H D_H). \quad (116)$$

These equations are the model-specific reduced perturbative cosmology used for the benchmark; they are not a replacement for super-horizon initial conditions or a Boltzmann hierarchy.

11.4. Reduced benchmark

We integrate Eqs. (115)–(116) from $z = 999$ on a flat reference background with

$$\Omega_{b0} = 0.049, \quad \Omega_{H0} = 0.251, \quad \Omega_{\lambda0} = 0.700, \quad (117)$$

using adiabatic growing-mode initial data and

$$\beta_H = 0.02, \quad m_{\lambda,f} = 5H_0, \quad c_H^2 = 10^{-9}, \quad \sigma_H = 0. \quad (118)$$

These are illustrative closure inputs, not values derived from soliton production. In particular, $m_{\lambda,f} = 5H_0$ is not treated as a generic heavy-field decoupling limit; the scalar is retained in the reduced perturbation system. The assumed sound speed corresponds to a cold coarse-grained population and is used only inside the long-wavelength regime. Table 4 compares the final growth factors with the uncoupled, pressureless system on the same background. The scalar-mediated force remains below 8.0×10^{-4} over the benchmark range, and the integrated changes in both Hopfion and baryon growth remain below 0.23%. Fig. 5 shows the time dependence for $k = 0.1 h \text{ Mpc}^{-1}$; the present scale dependence is summarized by the ϵ_H column in Table 4. This benchmark closes the candidate-level variables requested by the reduced model and shows that a stable, weakly deformed linear regime exists. It does not predict the production-generated c_H^2 , c_π , shot noise, or isocurvature, and it does not evolve radiation multipoles, recombination, lensing, or the residual M -wave sector. Those ingredients are required before the system can be promoted from a reduced perturbative candidate to a viable Boltzmann model.

12. Scientific status and remaining viability gates

Table 5 separates established candidate-level results from assumptions and open viability calculations.

The completed candidate-level core is

$$\left| \begin{array}{l} \text{explicit EFT} + \text{stationary topology-preserving Hopf sector} \\ + \text{projected local stability} + \text{reduced dynamical overlap trigger} \\ + \text{stable trigger principal sector} + \text{reduced linear cosmology} \\ + w_H \simeq 0, w_\lambda \simeq -1. \end{array} \right| \quad (119)$$

Table 5

Status of the construction.

Statement	Status	Limitation
Unique quadratic internal STF map $M \mapsto Q$ $Q: Q = C_2^2 - 4C_3^2$ $S^5 \rightarrow S^2 \times S^2$ and $\pi_3 = \mathbb{Z} \oplus \mathbb{Z}$ Unit-Hopf stationary configuration	Analytic proof Exact identity Analytic Numerical, topology preserved	Unique only in the stated operator class Internal Euclidean relation Requires the activated fixed-radius branch Reduced heavy-normal-mode EFT; local multistart, not global uniqueness
Positive non-collective projected gap	Same-operator refinement plus resolved large-box IR bands	Finite lattice; coupled normal, gravity, and mediator modes open
Self-generated overlap transition	Numerical homogeneous benchmark	Prescribed matter dilution and constant H ; open-system work identity
Reduced trigger $K_{\text{red}}, G_{\text{red}} > 0$	Analytic matrices plus tolerance audit	Complete for evolved q - χ - λ proxy, not full matter-metric-six- M system
Reduced final-branch perturbation system	Analytic plus quasi-static benchmark	Illustrative closure inputs; production coefficients, residual waves, shot noise, and super-horizon evolution open
Pressureless Hopfion gas	Candidate limit	Production, abundance, and collision integral open
Metastable negative-pressure scalar	Barrier, curvature mass, and flat-space bounce	Gravitational decay, radiative hierarchy, and observed scale open
Observationally viable joint model	Not established	Requires the gates below

Parameter dependence. The qualitative construction occupies open analytic regions rather than a single algebraic point. The activated target requires $\alpha > 0$, $\mu(\lambda_v) > 0$, and $\zeta > 0$, with the heavy-normal-mode reduction increasingly controlled as $\alpha\zeta/Z_M^2$ and $\mu_v\zeta/Z_M^2$ grow. The positive metastable scalar branch exists throughout $2b^2/9 < ac < b^2/4$; for $b = c = 1$ this is $2/9 < a < 1/4$, not only $a = 0.24$. The reduced trigger is stable on the open interval $0 \leq y < 2$, while the final closure requires $m_{\chi,f}^2 > 0$, $m_{\lambda,f}^2 > 0$, and sufficiently small illustrative finite-stiffness response. The one-at-a-time $\pm 5\%$ trigger audit described in Section 9 verifies a local neighborhood while exposing a capture boundary. These results are evidence of local parameter continuity, not a global robustness scan or a radiative-naturalness argument. The stronger statement of a viable dark-matter/dark-energy model requires the following gates.

Gate I: unrestricted soliton sector and collisions. Repeat the stationary calculation in the full six-component M theory at finite α and μ , verify continuum and box convergence for $(1, 0)$, $(0, 1)$, $(1, 1)$, and opposite-charge configurations, calculate the full projected fluctuation operator including radial and imbalance fields, and simulate relativistic collisions. Outputs are M_H , R_H , annihilation and transfer cross sections, charge exchange, radiation, and the hyperbolicity domain.

Gate II: expanding-lattice production. Evolve M_{ab} , χ , and λ in three dimensions on FLRW backgrounds. Determine the charge distribution, number density, correlation length, residual wave fraction, and complete energy budget. The pass criterion is an open parameter region that produces the observed dark-matter density after redshifting, rather than a hand-normalized point.

Gate III: astrophysical darkness. Use collision data to calculate $\sigma_{\text{self}}(v)/M_H$, annihilation, capture, scalar-mediated forces, equivalence-principle effects, stellar energy loss, and visible-sector scattering over dwarf-to-cluster velocities.

Gate IV: Boltzmann-complete perturbations. Extend the reduced final-branch system in Section 11 using the phase-space distribution produced by the expanding transition. This requires calculated sound speed and anisotropic stress, shot noise, compensated and uncompensated isocurvature, residual Goldstone and radial M waves, mediator excitations, super-horizon initial conditions, radiation and neutrino hierarchies, and the complete metric response. The reduced closure demonstrates a stable weak-coupling window but does not determine these production-dependent inputs.

Gate V: likelihood analysis. Implement the Boltzmann-complete system and confront CMB temperature, polarization and lensing, baryon acoustic oscillations, supernovae, growth, weak lensing, and small-scale structure [15–18]. A viable model requires a stable posterior region with acceptable spectra and no excluded early dark energy or isocurvature. The construction should be rejected or substantially reformulated if the unrestricted Hopf sector disappears, if the transition produces no conserved soliton abundance at continuum resolution, if required self-interactions violate astrophysical limits, if the false vacuum is too short-lived or radiatively unstable, or if the closed perturbation system contains a ghost, gradient instability, or excessive isocurvature.

13. Discussion

The model makes three precise advances over a purely phenomenological tensor insertion. First, the Fock connection carries algebraic and topological information: $L \cdot A = 0$ is the Pfaffian condition that maximizes the unique quadratic STF channel and selects a vacuum with Hopf sectors. Second, the spacetime tensor in the portal is the trace-free Hilbert stress of the stated M action, not an arbitrary reduction of internal indices; its variational response and total conservation law are explicit. Third, the dark-matter carrier and trigger have been tested beyond homogeneous invariant curvatures: a topology-preserving stationary unit-Hopf configuration and its collective-classified, refinement-tested projected Hessian have been computed, the overlap threshold has been crossed in a

self-generated homogeneous trajectory that remains in the reduced principal-stability window, and the final branch admits a closed reduced Newtonian-gauge consistency test with sub-percent deformations of its uncoupled reference. The construction also has clear limitations. Fock symmetry does not determine the EFT coefficients, the relic abundance, or the dark-energy scale. The mediator resolves the dangerous direct squared-overlap operator but introduces its own mass and couplings. The stationary calculation is a controlled low-energy Faddeev–Skyrme solve rather than a full finite-stiffness six-component scan. The trigger benchmark uses a prescribed matter bath and does not establish a complete cosmological history. The reduced perturbative closure assumes a dilute long-wavelength Hopfion fluid and parameterizes, rather than derives, its finite-stiffness scalar response, sound speed, and anisotropic stress. The massless tangent modes and finite-core processes can affect both cosmology and self-interactions. These limitations are why the present terminology is *topological dark-matter candidate plus metastable dark-energy candidate*, not an observationally viable replacement for Λ CDM. A natural model-building branch is to separate the topology trigger from the ultra-small vacuum-energy carrier. A heavier field could control μ , while a shift-protected lighter field supplies the long-lived vacuum branch. Such an extension may improve radiative hierarchy but adds degrees of freedom and must repeat the stability, production, and perturbation analyses.

14. Conclusion

Completed results. The hydrogenic Pfaffian condition selects decomposable $SO(4)$ bivectors, maximizes the norm of the unique lowest-order quadratic STF composite, and changes the fixed-radius vacuum to $S^2 \times S^2$. The mediator-resolved action now has an explicit field/operator census, portal variation, sector-exchange law, and reduced principal matrices. In the activated branch, a topology-preserving three-dimensional minimization produces the $(N_+, N_-) = (1, 0)$ stationary sector. Deterministic local multistarts recover the same basin, while collective-mode classification and same-operator refinement give a positive first non-collective gap with h^2 intercept 0.6781839719; resolved enlarged-box infrared bands contain no negative mode. The homogeneous proxy generates the overlap, crosses both thresholds, passes tolerance and local parameter-sensitivity checks, and remains inside the reduced principal-stability window. The positive scalar branch has an explicit barrier, curvature mass, and flat-space bounce coefficient $\hat{B}_4 = 1082.94756088$ with the scaling in Eq. (50).

Conditional interpretations. These results support a topological dark-matter candidate because the EFT contains a localized unit-charge carrier with tested local stability and a conditional dilute nonrelativistic $w_H \simeq 0$ limit. They support an overlap-mediated metastable dark-energy candidate because the same transition accesses a positive locally stable branch with $w_\lambda \simeq -1$ and calculable semiclassical suppression. The illustrative reduced Newtonian-gauge closure demonstrates a stable weak-coupling regime with sub-percent linear-growth deformations relative to its uncoupled reference; it is not an observational fit. The β_H , c_H^2 , and σ_H benchmark values are closure choices, not microscopic predictions.

Remaining viability gates. The revision does not establish relic production or abundance, acceptable soliton collisions and self-interactions, a minimum-energy finite-stiffness unwinding saddle, a gravitational Coleman–De Luccia lifetime, radiative naturalness, production-derived perturbation coefficients, Boltzmann-complete evolution, or likelihood viability. The decisive next calculations are unrestricted finite-stiffness soliton and scattering simulations, an expanding-lattice transition, a production-matched Boltzmann system, and a joint observational analysis. The present result is therefore a technically strengthened candidate theory, not a completed viable dark-sector model.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The accompanying reproducibility archive contains the topology-preserving spectral minimizer, augmented-Lagrangian solver, projected-Hessian implementation, collective-mode and enlarged-box Hessian audits, stationary multistart checkpoints, homogeneous trigger integrator with tolerance and sensitivity audits, flat-space bounce solver, reduced perturbation benchmark, numerical output files, and scripts used to generate the figures and numerical values reported here.

Data availability

No data was used for the research described in the article.

Appendix A. Component verification of the Fock identities

For

$$M = \begin{pmatrix} 0 & L_3 & -L_2 & A_1 \\ -L_3 & 0 & L_1 & A_2 \\ L_2 & -L_1 & 0 & A_3 \\ -A_1 & -A_2 & -A_3 & 0 \end{pmatrix}, \quad (\text{A.1})$$

direct calculation gives

$$C_1 = \mathbf{L}^2 + \mathbf{A}^2, \quad \text{Pf } M = \mathbf{L} \cdot \mathbf{A}, \quad \text{Tr}(M M^T) = 2C_1, \quad (\text{A.2})$$

and

$$\text{Tr}(Q^2) = C_1^2 - 4(\mathbf{L} \cdot \mathbf{A})^2, \quad Q^2 - \frac{1}{4}[C_1^2 - 4(\mathbf{L} \cdot \mathbf{A})^2]\mathbf{1} = 0. \quad (\text{A.3})$$

A symbolic verification can be generated directly from these component definitions.

Appendix B. Derivation of the perfect-fluid principal coefficients

For a canonically normalized scalar-like order-parameter fluctuation,

$$S_{\mu\nu} = \partial_\mu q \partial_\nu q - \frac{1}{4} g_{\mu\nu} (\partial q)^2. \quad (\text{B.1})$$

Contracting with $T_m^{\mu\nu} = (\rho + p)u^\mu u^\nu + p g^{\mu\nu}$ gives

$$S_{\mu\nu} T_m^{\mu\nu} = (\rho + p)(u \cdot \partial q)^2 + \left(p - \frac{1}{4}T\right)(\partial q)^2 \quad (\text{B.2})$$

$$= (\rho + p) \left[(u \cdot \partial q)^2 + \frac{1}{4}(\partial q)^2 \right]. \quad (\text{B.3})$$

In the rest frame, $(\partial q)^2 = -\dot{q}^2 + (\nabla q)^2$, yielding [Eq. \(53\)](#). Adding $g\chi^2 S : T_m / \Lambda^6$ to the canonical quadratic scalar Lagrangian gives [Eq. \(55\)](#). For $y \geq 0$, K_M is positive; $G_M > 0$ gives $y < 2$; and $G_M \leq K_M$ gives $c_M^2 \leq 1$.

Appendix C. Quartic portal variation

For one duality sector write

$$\mathcal{L}_{4,s} = c(A_{\mu\nu} A^{\mu\nu} - A^2), \quad c = \frac{\zeta}{16\sigma_0^4}, \quad A_{\mu\nu} = \nabla_\mu M_s \cdot \nabla_\nu M_s, \quad A = A_\mu{}^\mu. \quad (\text{C.1})$$

Its Hilbert tensor is

$$\Theta_{\alpha\beta}^{(4,s)} = c g_{\alpha\beta} (A_{\mu\nu} A^{\mu\nu} - A^2) - 4c (A_\alpha{}^\rho A_{\beta\rho} - A A_{\alpha\beta}). \quad (\text{C.2})$$

In four spacetime dimensions $\Theta^{(4,s)} = 0$, so this contribution is already trace-free. Define

$$\mathcal{P} = T_m^{\mu\nu} A_{\mu\nu}, \quad \mathcal{Q}_T = T_m^{\alpha\beta} A_\alpha{}^\rho A_{\beta\rho}. \quad (\text{C.3})$$

Then

$$X_{4,s} = c [T_m (A_{\mu\nu} A^{\mu\nu} - A^2) - 4\mathcal{Q}_T + 4A\mathcal{P}] \quad (\text{C.4})$$

and differentiation with respect to $A_{\kappa\lambda}$ gives

$$\begin{aligned} C_s^{\kappa\lambda} = c \Big\{ & 2T_m (A^{\kappa\lambda} - A g^{\kappa\lambda}) - 4(T_m^{\kappa\beta} A_{\beta}{}^\lambda + T_m^{\lambda\beta} A_{\beta}{}^\kappa) \\ & + 4(\mathcal{P} g^{\kappa\lambda} + A T_m^{\kappa\lambda}) \Big\}. \end{aligned} \quad (\text{C.5})$$

The quartic portal contribution is therefore

$$\mathcal{E}_{M_s^I}^{\text{P},4} = -2\nabla_\kappa [F(\chi) C_s^{\kappa\nu} \nabla_\nu M_s^I]. \quad (\text{C.6})$$

It contains at most second derivatives of M_s^I , as stated in [Section 3.3](#).

Appendix D. Projected-Hessian construction

Let $\mathbf{u}$ collect the two tangent coordinates at all interior sites. At the constrained stationary point the relevant second variation is

$$H_c = \frac{\partial^2}{\partial \mathbf{u}^2} (E - \alpha Q_H) \Big|_{\mathbf{u}=0}. \quad (\text{D.1})$$

With $g_Q = \nabla_{\mathbf{u}} Q_H$, define

$$P_Q = I - \frac{g_Q g_Q^T}{g_Q^T g_Q}. \quad (\text{D.2})$$

The raw reported operator is $P_Q H_c P_Q$, using the unweighted lattice tangent inner product. Hessian-vector products are obtained by nested automatic differentiation without storing a dense 6750×6750 matrix. The charge projection creates a projector-null direction rather than a physical fluctuation mode. For the classified spectrum we additionally project the orthonormalized span of the target rotation about the fixed vacuum axis, three translations, and three spatial rotations. Finite volume and the fixed boundary turn the latter directions into pseudozeros; they are kept in the raw spectrum of [Eq. \(74\)](#) and removed only for the non-collective refinement in [Table 3](#).

Appendix E. Numerical reproducibility fields

Each stationary output stores the full $N \times N \times N \times 3$ field and JSON metadata containing N , L , h , E_2 , E_4 , Q_H , virial mismatch, multiplier history, and constrained residual. The Hessian audit stores the input configuration, projected and collective-deflated dimensions, collective Gram data, eigenvalues in site and physical normalizations, eigenpair residuals, solver status, and tolerances. The trigger output stores the complete time series for q , $\dot{q}$, χ , $\dot{\chi}$, λ , $\dot{\lambda}$, ρ , X , K_M , G_M , c_M^2 , $m_{\lambda,\text{eff}}^2$, and the open-system work-balance residual. Separate JSON artifacts archive the Hessian refinement and large-box bands, trigger tolerance and one-at-a-time sensitivity runs, stationary multistarts, and the flat-space bounce with virial and truncation checks.

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